



Peptic Ulcer Disease

By [Nimish Vakil](#), MD, University of Wisconsin School of Medicine and Public Health

Reviewed By [Minhhuyen Nguyen](#), MD, Fox Chase Cancer Center, Temple University

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A peptic ulcer is an erosion in a segment of the gastrointestinal mucosa, typically in the stomach (gastric ulcer) or the first few centimeters of the duodenum (duodenal ulcer), that penetrates through the muscularis mucosae. Nearly all ulcers are caused by [Helicobacter pylori infection](#) or nonsteroidal anti-inflammatory drug (NSAID) use. Symptoms typically include burning epigastric pain that is often relieved by food. Diagnosis is by endoscopy and testing for *Helicobacter pylori*. Treatment involves acid suppression, eradication of *H. pylori* (if present), and avoidance of NSAIDs.

(See also [Overview of Acid Secretion](#) and [Overview of Gastritis](#).)

Ulcers may range in size from several millimeters to several centimeters. Ulcers are delineated from erosions by the depth of penetration; erosions are more superficial and do not involve the muscularis mucosae.

Ulcers can occur at any age, including infancy and childhood, but are most common among middle-aged adults.

Etiology of Peptic Ulcer Disease

H. pylori and nonsteroidal anti-inflammatory drugs (NSAIDs) disrupt normal mucosal defense and repair, making the mucosa more susceptible to acid. [H. pylori infection](#) is present in over 50% of patients with duodenal ulcers (1) and in 30 to 50% of patients with gastric ulcers. If *H. pylori* is eradicated, only 10% of patients have recurrence of peptic ulcer disease (2), compared with 70% recurrence in patients treated with acid suppression alone. NSAIDs now account for > 50% of peptic ulcers.

Cigarette smoking is a risk factor for the development of ulcers and their complications (3). Also, smoking impairs ulcer healing and increases the incidence of recurrence. Risk correlates with the number of cigarettes smoked per day. Although alcohol is a strong promoter of acid secretion, no definitive data link moderate amounts of alcohol to the development or delayed healing of ulcers. Very few patients have hypersecretion of gastrin caused by a [gastrinoma](#) (Zollinger-Ellison syndrome).

Etiology references

1. [Ciociola AA, McSorley DJ, Turner K, Sykes D, Palmer JB](#): *Helicobacter pylori* infection rates in duodenal ulcer patients in the United States may be lower than previously estimated. *Am J Gastroenterol* 94(7):1834–1840, 1999. doi:10.1111/j.1572-0241.1999.01214.x
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3. [Ostensen H, Gudmundsen TE, Ostensen M, Burhol PG, Bonnevie O](#): Smoking, alcohol, coffee, and familial factors: any associations with peptic ulcer disease? A clinically and radiologically prospective study. *Scand J Gastroenterol* 20(10):1227–1235, 1985. doi:10.3109/00365528509089281

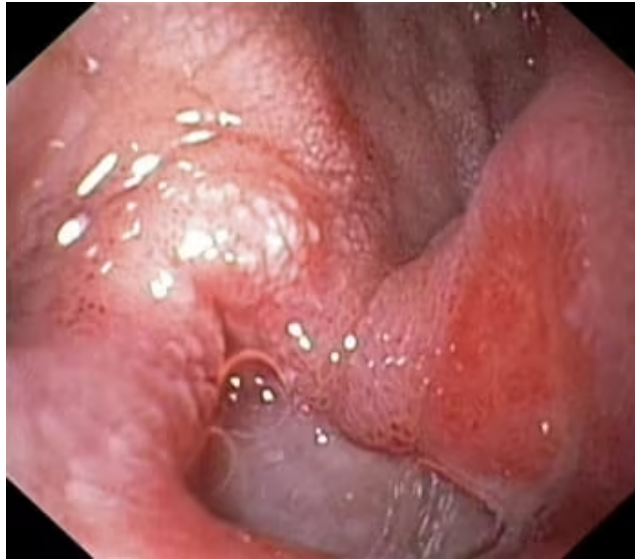
Symptoms and Signs of Peptic Ulcer Disease

Symptoms of peptic ulcer disease depend on ulcer location and patient age; many patients, particularly older patients, have few or no symptoms. Pain is most common, often localized to the epigastrium and relieved by food or antacids. The pain is described as burning or gnawing, or sometimes as a sensation of hunger. The course is usually chronic and recurrent. Only about half of patients present with the characteristic pattern of symptoms.

Gastric ulcer symptoms often do not follow a consistent pattern (eg, eating sometimes exacerbates rather than relieves pain). This is especially true for pyloric channel ulcers, which are often associated with symptoms of obstruction (eg, bloating, nausea, vomiting) caused by edema and scarring.

Duodenal ulcers tend to cause more consistent pain. Pain is absent when the patient awakens but appears mid-morning and is relieved by food but recurs 2 to 3 hours after a meal. Pain that awakens a patient at night is common and is highly suggestive of duodenal ulcer. In neonates, perforation and hemorrhage may be the first manifestation of duodenal ulcer. Hemorrhage may also be the first recognized sign in later infancy and early childhood, although repeated vomiting or evidence of abdominal pain may be a clue.

Duodenal Ulcer



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A large duodenal ulcer is seen at the 6 o'clock position.

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Diagnosis of Peptic Ulcer Disease

- Endoscopy
- Sometimes serum gastrin levels

Diagnosis of peptic ulcer is suggested by patient history and confirmed by endoscopy. Empiric therapy is often begun without definitive diagnosis. However, endoscopy allows for biopsy or cytologic brushing of gastric and esophageal lesions to distinguish between simple ulceration and ulcerating stomach cancer. [Stomach cancer](#) may manifest with similar manifestations and must be excluded, especially in patients who are > 60 years, have lost weight, or report severe or refractory symptoms. The incidence of malignant duodenal ulcer is extremely low, so biopsies of duodenal lesions are generally not warranted. Endoscopy can also be used to definitively diagnose *H. pylori* infection, which should be sought when an ulcer is detected (see [diagnosis of H. pylori infection](#)).

Gastric Ulcer

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A large gastric ulcer crater is visible at the center of the image. It is important to biopsy the edges of the ulcer because of the risk of cancer.

Image provided by David M. Martin, MD.

Gastrin-secreting cancer and gastrinoma should be considered when there are multiple ulcers, when ulcers develop in atypical locations (eg, postbulbar) or are refractory to treatment, or when the patient has prominent diarrhea or weight loss. Serum gastrin levels should be measured in these patients.

Complications of Peptic Ulcer Disease

Hemorrhage

Mild to severe [gastrointestinal hemorrhage](#) is the most common complication of peptic ulcer disease.

Symptoms include hematemesis (vomiting of fresh blood or coffee-ground material), passage of bloody stools (hematochezia) or black tarry stools (melena), and weakness, orthostasis, syncope, thirst, and sweating caused by blood loss.

Penetration (confined perforation)

A peptic ulcer may penetrate the wall of the stomach or duodenum. Perforation occurs when the ulcer base erodes completely leaving an opening that allows gastric contents to leak into the peritoneum. Chronic ulcers may cause extensive adhesions and, in this circumstance, a perforated ulcer may leak gastric contents into an area circumscribed by adhesions (confined perforation). The ulcer may penetrate into the adjacent confined space (lesser sac) or another organ (eg, pancreas, liver).

Pain may be intense, persistent, referred to sites other than the abdomen (usually the back when caused by penetration of a posterior duodenal ulcer into the pancreas), and modified by body position.

CT or MRI is usually needed to confirm the diagnosis.

When therapy does not result in healing, surgery is required.

Gastric Ulcer Perforation



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This image shows chronic, confined, gastric ulcer perforation through which the liver is visible.

Image provided by David M. Martin, MD.

Free perforation

Ulcers that perforate into the peritoneal cavity unchecked by adhesions are usually located in the anterior wall of the duodenum or, less commonly, in the stomach.

The patient presents with an acute abdomen. There is sudden, intense, continuous epigastric pain that spreads rapidly throughout the abdomen, often becoming prominent in the right lower quadrant and at times referred to one or both shoulders. The patient usually lies still because even deep breathing worsens the pain. Palpation of the abdomen is painful, rebound tenderness is prominent, abdominal muscles are rigid (boardlike), and bowel sounds are diminished or absent. Shock may ensue, heralded by increased pulse rate and decreased blood pressure and urine output. Symptoms may be less striking in patients who are older or moribund and in those receiving corticosteroids or immunosuppressants.

Free Air



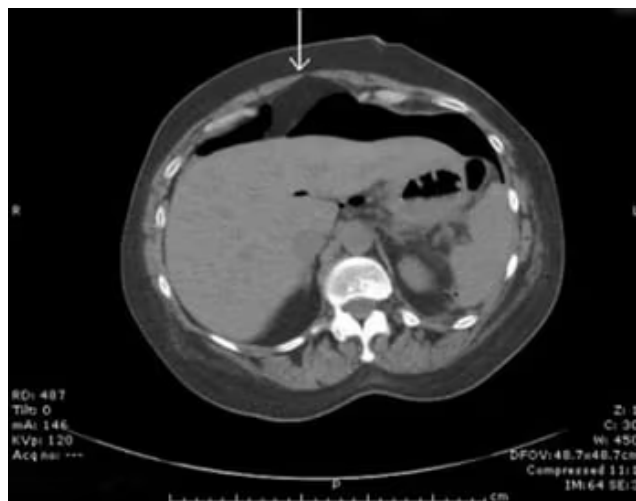
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This abdominal radiograph shows free air caused by a perforation.

Image provided by Parswa Ansari, MD.

Diagnosis of a free perforation is confirmed if a radiograph or CT scan shows free air under the diaphragm or in the peritoneal cavity. Upright views of the chest and abdomen are preferred. The most sensitive view is the lateral radiograph of the chest. Severely ill patients may be unable to sit upright and should have a lateral decubitus radiograph of the abdomen. Failure to detect free air does not exclude the diagnosis.

CT With Free Peritoneal Air



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Free air is seen anterior to the liver. The arrow points to the falciform ligament.

Image provided by Parswa Ansari, MD.

Immediate surgery is required. The longer the delay, the poorer is the prognosis. IV antibiotics effective against intestinal flora (eg, cefotetan, or amikacin plus clindamycin) should be given. Usually, a nasogastric tube is inserted to do continuous nasogastric suction. In the rare cases when surgery cannot be done, prognosis is poor.

Gastric outlet obstruction

Obstruction may be caused by scarring, spasm, or inflammation resulting from an ulcer.

Symptoms include recurrent, large-volume vomiting, occurring more frequently at the end of the day and often as late as 6 hours after the last meal. Loss of appetite with persistent bloating or fullness after eating also suggests gastric outlet obstruction. Prolonged vomiting may cause weight loss, dehydration, and alkalosis.

If the patient's history suggests obstruction, physical examination, gastric aspiration, or radiographs may provide evidence of retained gastric contents. A succussion splash heard > 6 hours after a meal or aspiration of fluid or food residue > 200 mL after an overnight fast suggests gastric retention. If gastric aspiration shows marked retention, the stomach should be emptied and endoscopy or cross-sectional imaging with CT should be done to determine site, cause, and degree of obstruction.

Edema or spasm caused by an active pyloric channel ulcer is treated with gastric decompression by nasogastric suction and acid suppression (eg, IV histamine-2 receptor antagonist or IV proton pump inhibitor). Dehydration and electrolyte imbalances resulting from protracted vomiting or continued nasogastric suctioning should be vigorously sought and corrected. Prokinetic agents are not indicated. Generally, obstruction resolves within 2 to 5 days of treatment. Prolonged obstruction may result from

peptic scarring and may respond to endoscopic pyloric balloon dilation. Surgery is necessary to relieve obstruction in selected cases.

Recurrence

Factors that affect recurrence of ulcer include failure to eradicate *H. pylori*, continued nonsteroidal anti-inflammatory drug (NSAID) use, and smoking. Less commonly, a gastrinoma may be the cause. The 3-year recurrence rate for gastric and duodenal ulcers is < 10% when *H. pylori* is successfully eradicated ([1](#), [2](#)) but is > 50% when it is not. Thus, a patient with recurrent disease should be [tested for *H. pylori*](#) and treated again if the tests are positive.

Although long-term treatment with histamine-2 receptor antagonists, proton pump inhibitors, or misoprostol reduces the risk of recurrence, the routine use of these medications for this purpose is not recommended. However, patients who require NSAIDs after having had a peptic ulcer are candidates for long-term therapy, as are those with a marginal ulcer or prior perforation or bleeding.

Stomach cancer

Patients with *H. pylori*-associated ulcers have a 3- to 6-fold increased risk of [gastric cancer](#) later in life. Eradication of the organism is therefore important both to prevent ulcer recurrence and subsequent cancer. There is no increased risk of cancer with ulcers of other etiology.

Complications references

1. [Ford AC, Gurusamy KS, Delaney B, Forman D, Moayyedi P](#): Eradication therapy for peptic ulcer disease in *Helicobacter pylori*-positive people. *Cochrane Database Syst Rev* 4(4):CD003840, 2016. doi:10.1002/14651858.CD003840.pub5
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Treatment of Peptic Ulcer Disease

- Eradication of *H. pylori* (when present)
- Acid-suppressive medications

Treatment of gastric and duodenal ulcers requires [eradication of *H. pylori*](#) when present and a reduction of gastric acidity ([1](#)). For duodenal ulcers, it is particularly important to suppress nocturnal acid secretion.

Methods of decreasing acidity include a number of medications. In addition, mucosal-protective medications (eg, sucralfate) and acid-reducing surgical procedures may be used. [Medications](#) are discussed elsewhere.

Adjuncts

Smoking should be stopped, and alcohol consumption should be stopped or limited to small amounts of dilute alcohol. There is no evidence that changing the diet speeds ulcer healing or prevents recurrence. Thus, many physicians recommend eliminating only foods that cause distress.

Surgery

With current medications, the number of patients requiring surgery has declined dramatically (2). Indications include perforation, obstruction, uncontrolled or recurrent bleeding, and, although rare, symptoms that do not respond to medication.

Surgery consists of a procedure to reduce acid secretion, often combined with a procedure to ensure gastric drainage. The recommended operation for duodenal ulcer is highly selective, or parietal cell, vagotomy (which is limited to nerves at the gastric body and spares antral innervation, thereby obviating the need for a drainage procedure). This procedure has a very low mortality rate and avoids the morbidity associated with resection and traditional vagotomy. Other acid-reducing surgical procedures include antrectomy, hemigastrectomy, partial gastrectomy, and subtotal gastrectomy (ie, resection of 30 to 90% of the distal stomach). These are typically combined with truncal vagotomy. Patients who undergo a resective procedure or who have an obstruction require gastric drainage via a gastroduodenostomy (Billroth I) or gastrojejunostomy (Billroth II).

The incidence and type of postsurgical symptoms vary with the type of operation. After resective surgery, up to 30% of patients have significant symptoms, including weight loss, maldigestion, anemia, dumping syndrome, reactive hypoglycemia, bilious vomiting, mechanical problems, and ulcer recurrence.

Weight loss is common after subtotal gastrectomy; the patient may limit food intake because of early satiety (because the residual gastric pouch is small) or to prevent dumping syndrome and other postprandial syndromes. With a small gastric pouch, distention or discomfort may occur after a meal of even moderate size; patients should be encouraged to eat smaller and more frequent meals.

Maldigestion and steatorrhea caused by pancreaticobiliary bypass, especially with Billroth II anastomosis, may contribute to weight loss.

Anemia is common (usually from [iron deficiency](#), but occasionally from [vitamin B12 deficiency](#) caused by loss of intrinsic factor or bacterial overgrowth in the afferent limb), and osteomalacia may occur. IM vitamin B12 supplementation is recommended for all patients with total gastrectomy but may also be given to patients with subtotal gastrectomy if deficiency is suspected.

Dumping syndrome may occur after gastric surgical procedures, particularly resections. Weakness, dizziness, sweating, nausea, vomiting, and palpitations occur soon after eating, especially hyperosmolar foods. This phenomenon is referred to as early dumping, the cause of which remains unclear but likely involves autonomic reflexes, intravascular volume contraction, and release of vasoactive peptides from the small intestine. Dietary modifications, with smaller, more frequent meals and decreased carbohydrate intake, usually help.

Reactive hypoglycemia or **late dumping** (another form of the syndrome) results from rapid emptying of carbohydrates from the gastric pouch. Early high peaks in blood glucose stimulate excess release of insulin, which leads to symptomatic hypoglycemia several hours after the meal. A high-protein, low-carbohydrate diet and adequate caloric intake (in frequent small feedings) are recommended.

Mechanical problems (including gastroparesis and [bezoar formation](#)) may occur secondary to a decrease in phase III gastric motor contractions, which are altered after antrectomy and vagotomy. Diarrhea is especially common after vagotomy, even without a resection (pyloroplasty).

Ulcer recurrence, is diagnosed by endoscopy and generally responds to either proton pump inhibitors or histamine-2 receptor antagonists. For ulcers that continue to recur, the completeness of vagotomy should be tested by gastric secretory studies, *H. pylori* should be eliminated if present, and gastrinoma should be ruled out by serum gastrin studies.

Treatment references

1. [Chey WD, Howden CW, Moss SF, et al](#): ACG Clinical Guideline: Treatment of *Helicobacter pylori* Infection. *Am J Gastroenterol* 119(9):1730–1753, 2024. doi: 10.14309/ajg.0000000000002968
2. [Masoudpour H, Wassef J, Saladziute S, Sherman J](#): Surgical Therapy of Gastric Ulcer Disease. *Surg Clin North Am* 105(1):173–186, 2025. doi:10.1016/j.suc.2024.06.013

Key Points

- Peptic ulcers affect the stomach or duodenum and can occur at any age, including infancy and childhood.
- Most ulcers are caused by *H. pylori* infection or nonsteroidal anti-inflammatory drug use; both factors disrupt normal mucosal defense and repair, making the mucosa more susceptible to acid.
- Burning pain is common; food may worsen gastric ulcer symptoms but relieve duodenal ulcer symptoms.
- Acute complications include gastrointestinal bleeding and perforation; chronic complications include gastric outlet obstruction, recurrence, and, when *H. pylori* infection is the cause, stomach cancer.
- Diagnose with endoscopy and do tests for *H. pylori* infection.
- Give acid-suppressing medications and antibiotics to eradicate *H. pylori*.